

KVFidelity Trace Atlas v4

Casey Reas intervention · CASK AIME24 n=30 · Qwen3-8B

A correct answer is not a score. It is a temporal event.

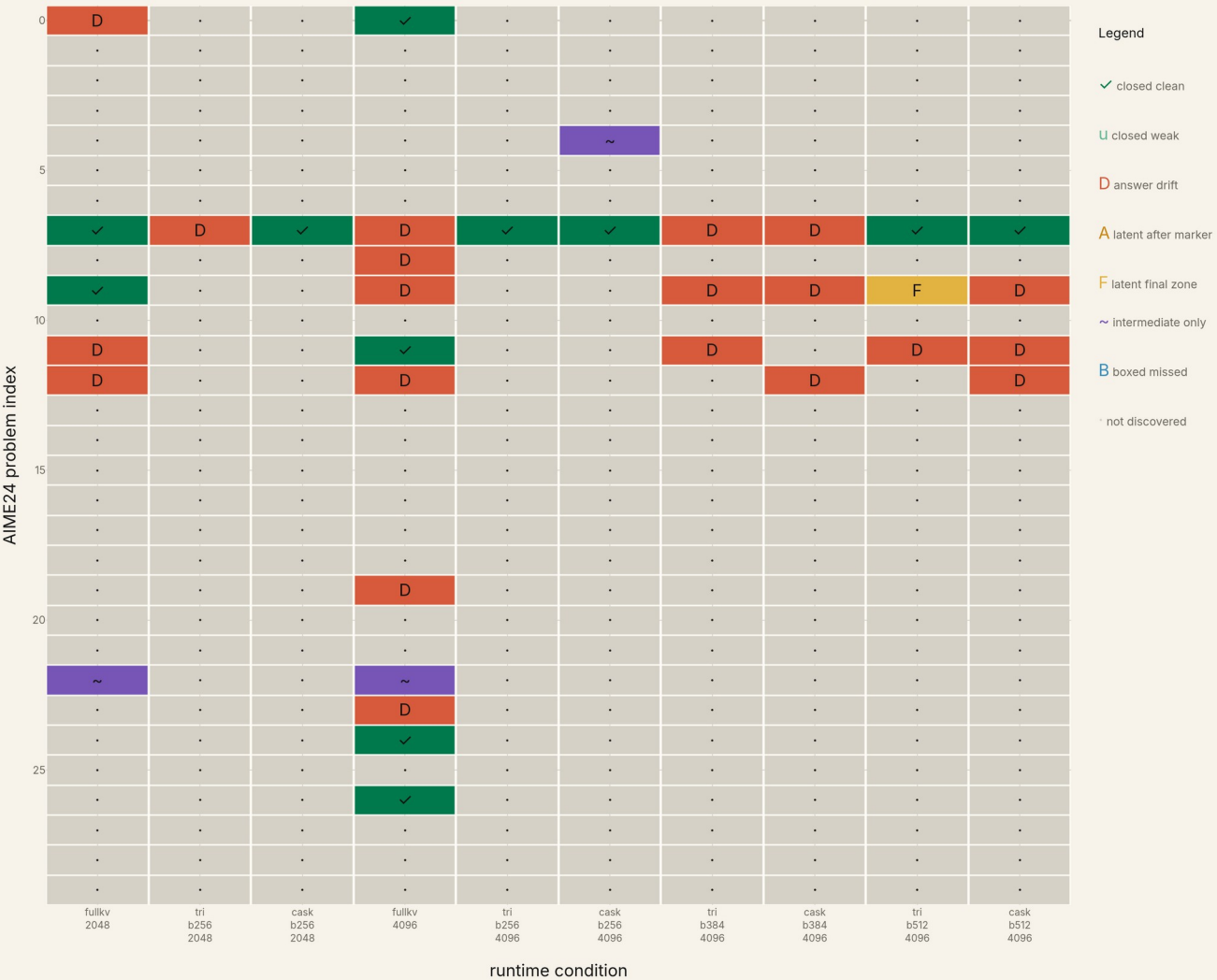
It emerges, persists, drifts, closes, or disappears.

CASK is the experiment. KVFidelity is the lens.

Main deck first. Defensive evidence moves to appendix.

Topology matrix

Topology matrix: answer as temporal event



Answer lifecycle

Node labels are counts out of 30; orange bands mark loss between lifecycle stages.

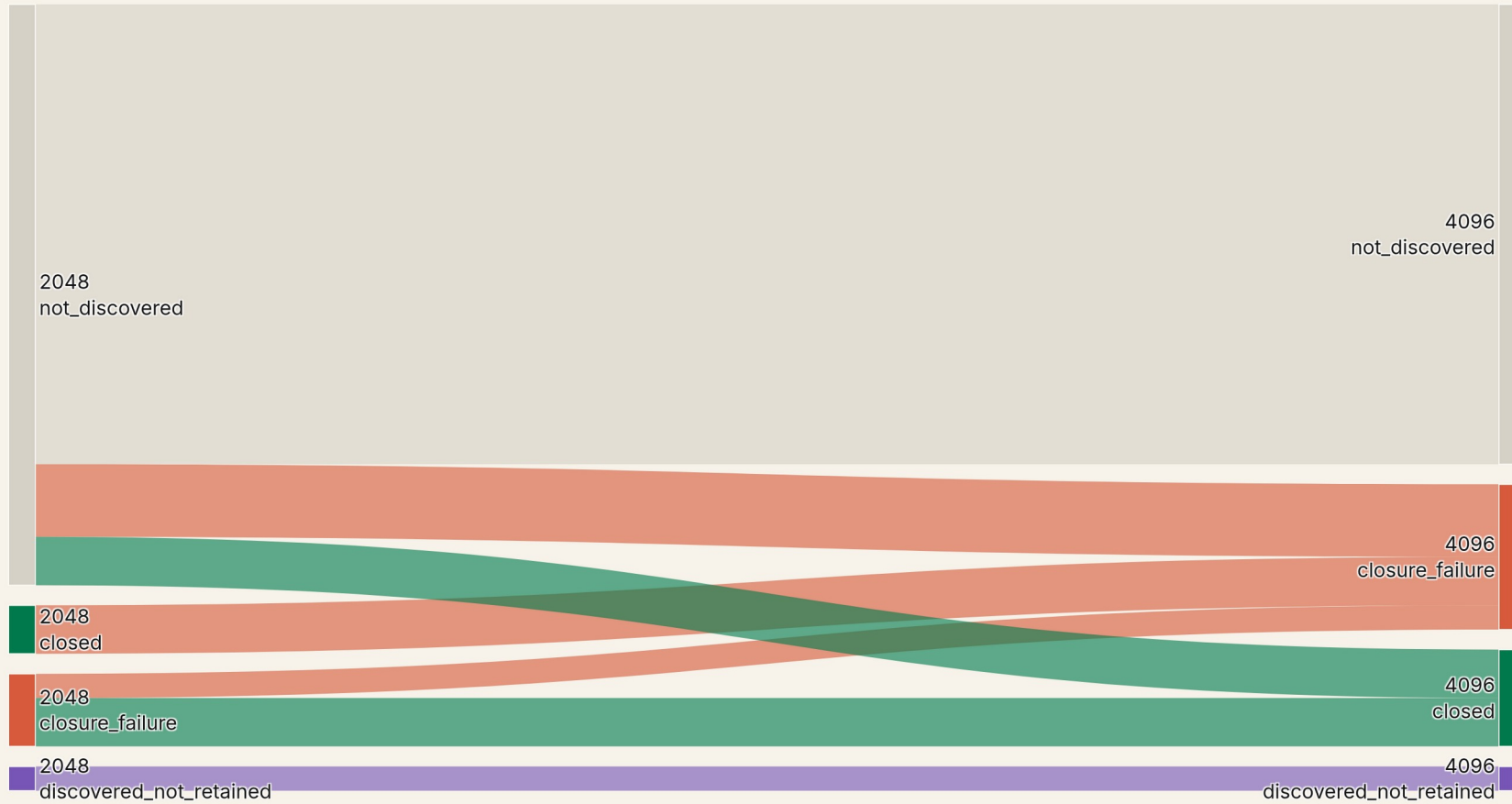
Answer lifecycle: where the answer is lost



FullKV 2048 → 4096

More generation transforms states: it repairs some samples and destabilizes others.

FullKV: answer states transform from 2048 → 4096



Case idx 7

Compression island: several compressed runs close cleanly while FullKV 4096 drifts.

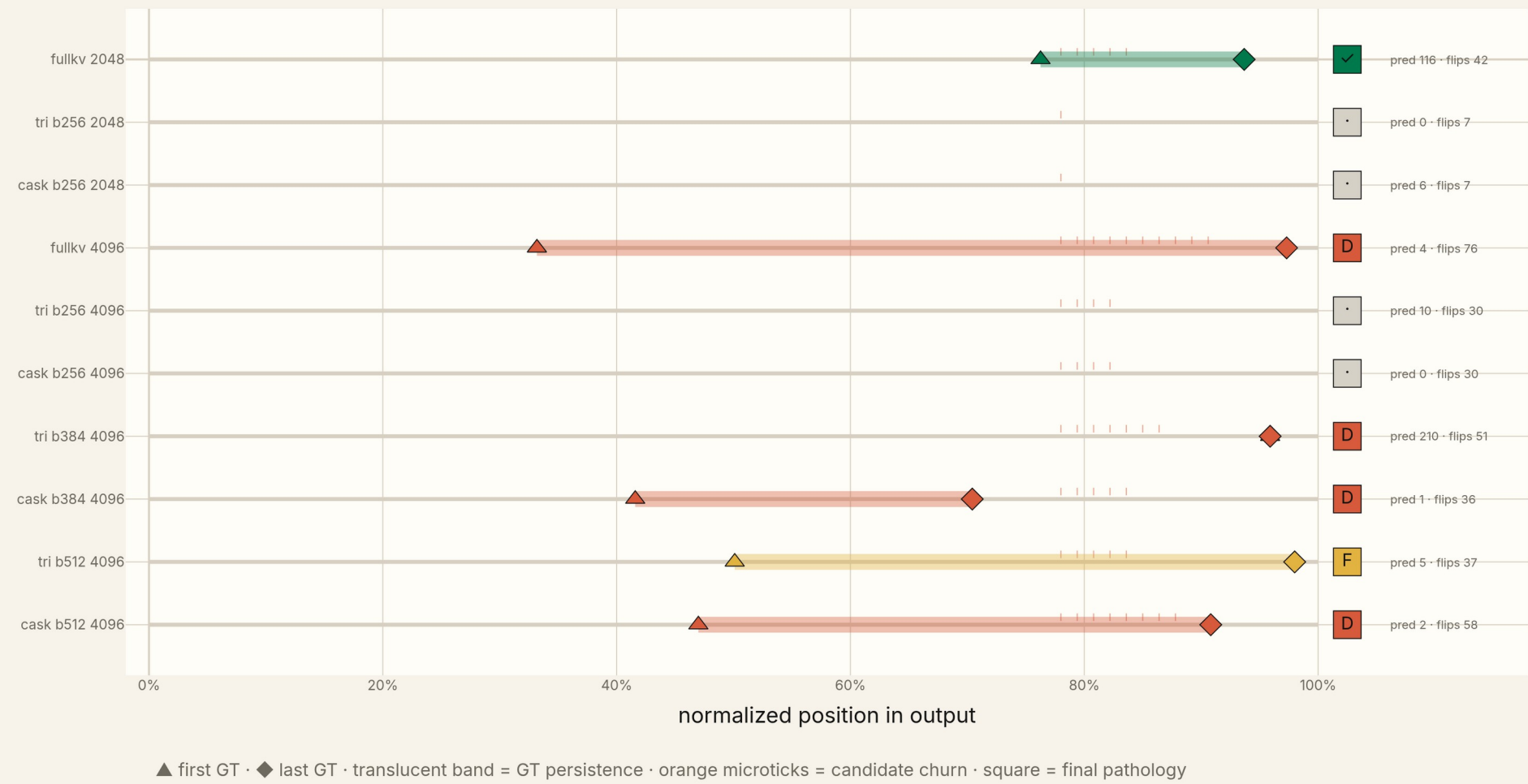
Case strip idx 7: GT 25



Case idx 9

The answer appears, persists, and still fails to close.

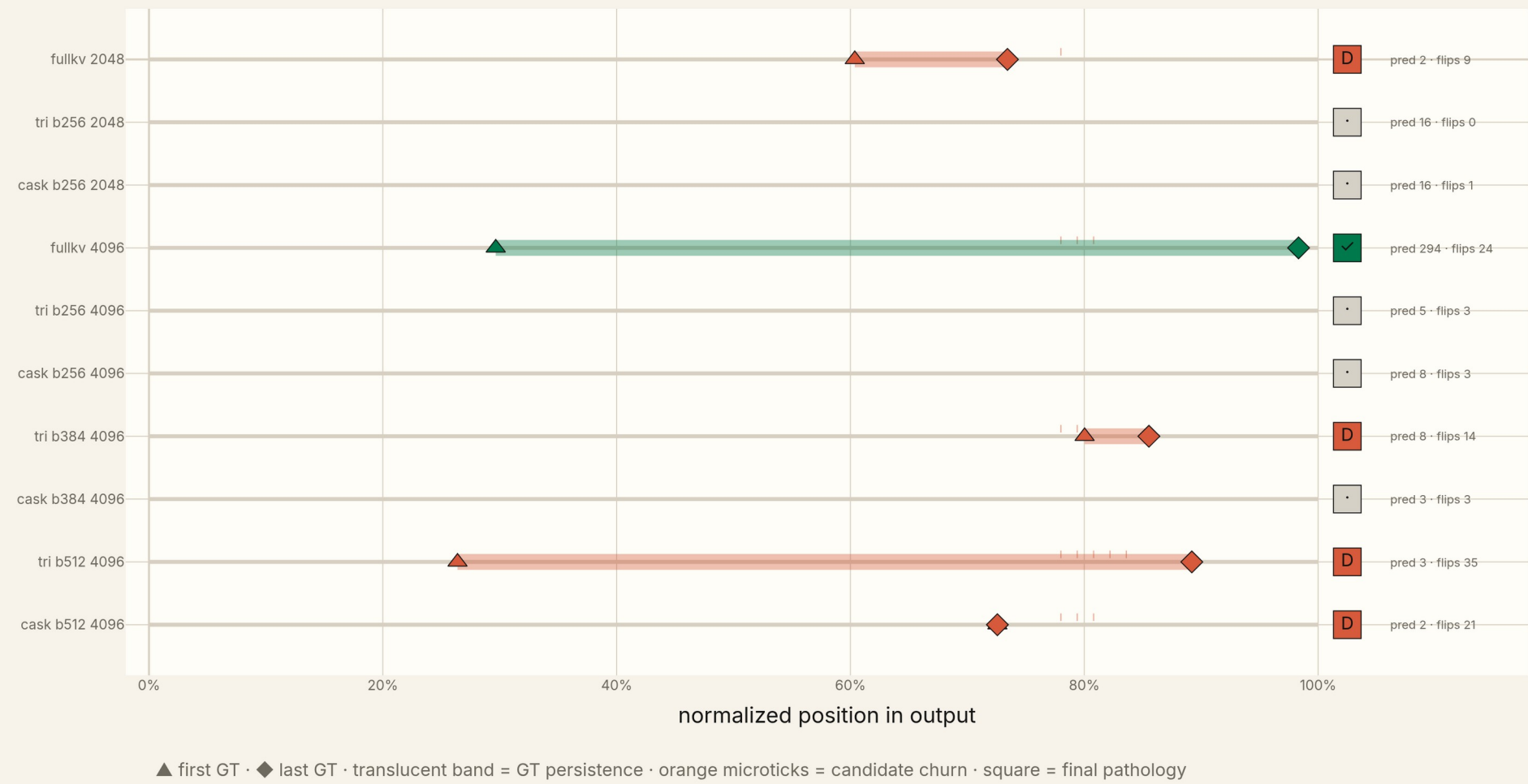
Case strip idx 9: GT 116



Case idx 11

More generation repairs FullKV closure; compression does not recover it here.

Case strip idx 11: GT 294



How the label is born

Make the measurement process visible: output excerpt → signals → pathology label.

How the label is born: idx 9 / FullKV 4096

The atlas is strongest when the measurement process becomes visible, not only the result.

Trace signals

run	fullkv_4096
GT	116
evaluator pred	4
pathology	answer drift
first GT frac	0.332
last GT frac	0.973
candidate flips	76
closure	failed

Visual grammar

- ▲ first GT
- ◆ last GT
- final label
- ▤ churn ticks

Output tail excerpt

```
* 23, and 1 has no common factors. So, the fraction is reduced, so m = 1, n = 115, so m + n
is [[116]]. Therefore, the answer is [[116]]. / / But let me check with another approach. Let
me think of it as probabilities step by step. Suppose that Jen has selected her four
numbers. Let me think of the probability that the lottery has exactly k matches with her
numbers, for k = 0,1,2,3,4. Then, the probability of winning a prize is  $P(k \geq 2) = P(2) +$ 
 $P(3) + P(4)$ . The probability of winning the grand prize is P(
```

Label: answer_marker_drift. The GT appears in answer-like regions, but the evaluator-visible final commitment drifts to 4.

Thesis

KVFidelity becomes strong when the correct answer stops being a score.

It becomes an event with a temporal shape.

The useful object is not a leaderboard. It is a score/partiture of answer trajectories.

Public claim remains deferred until manual label audit.

Appendix

Counts, caveats, alternate transitions, and full case strips.

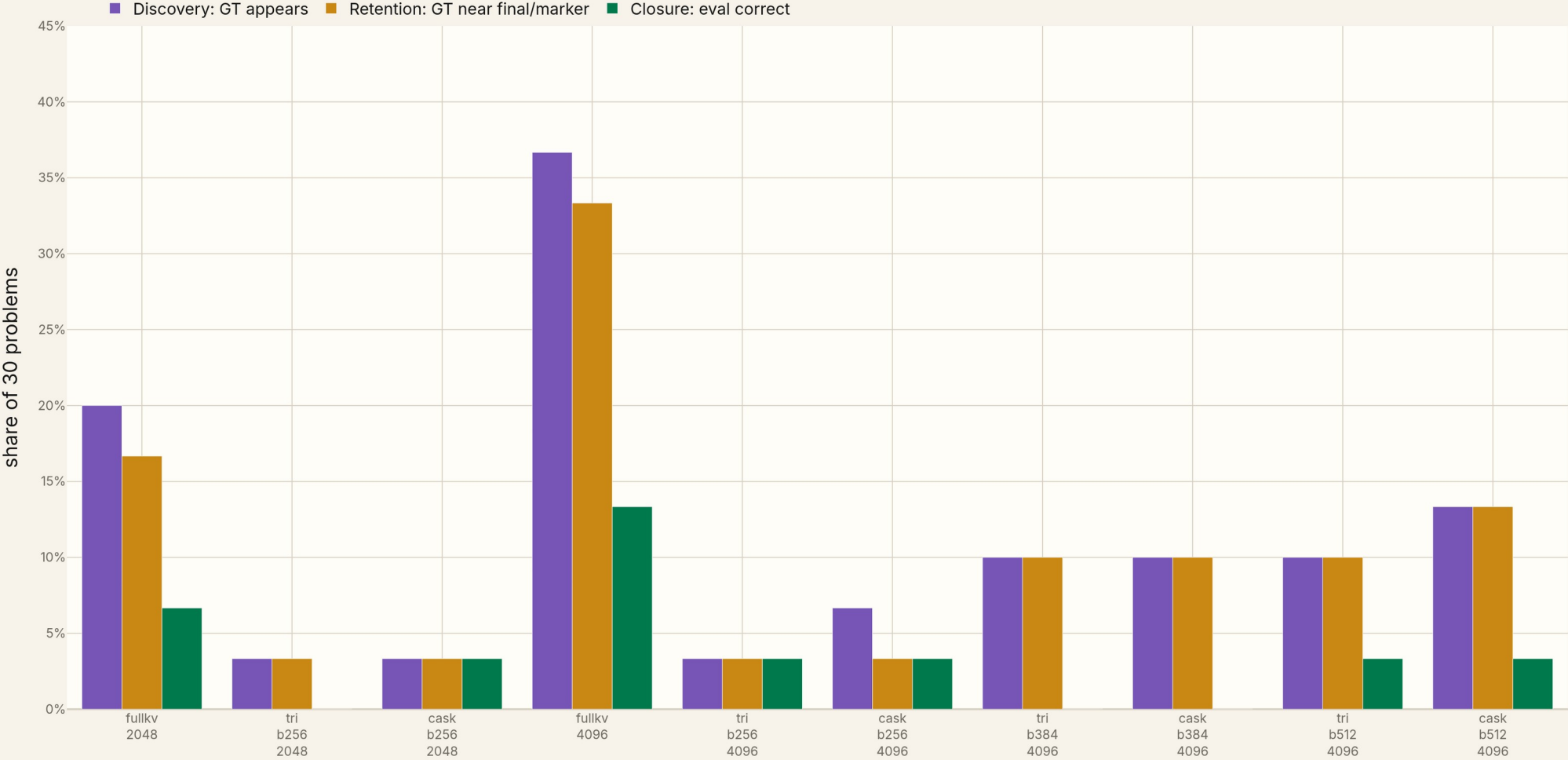
Appendix: evidence and defensive material

The main deck shows behavior. The appendix keeps the accounting.

Appendix · D/R/C bars

Conventional count view retained for auditability.

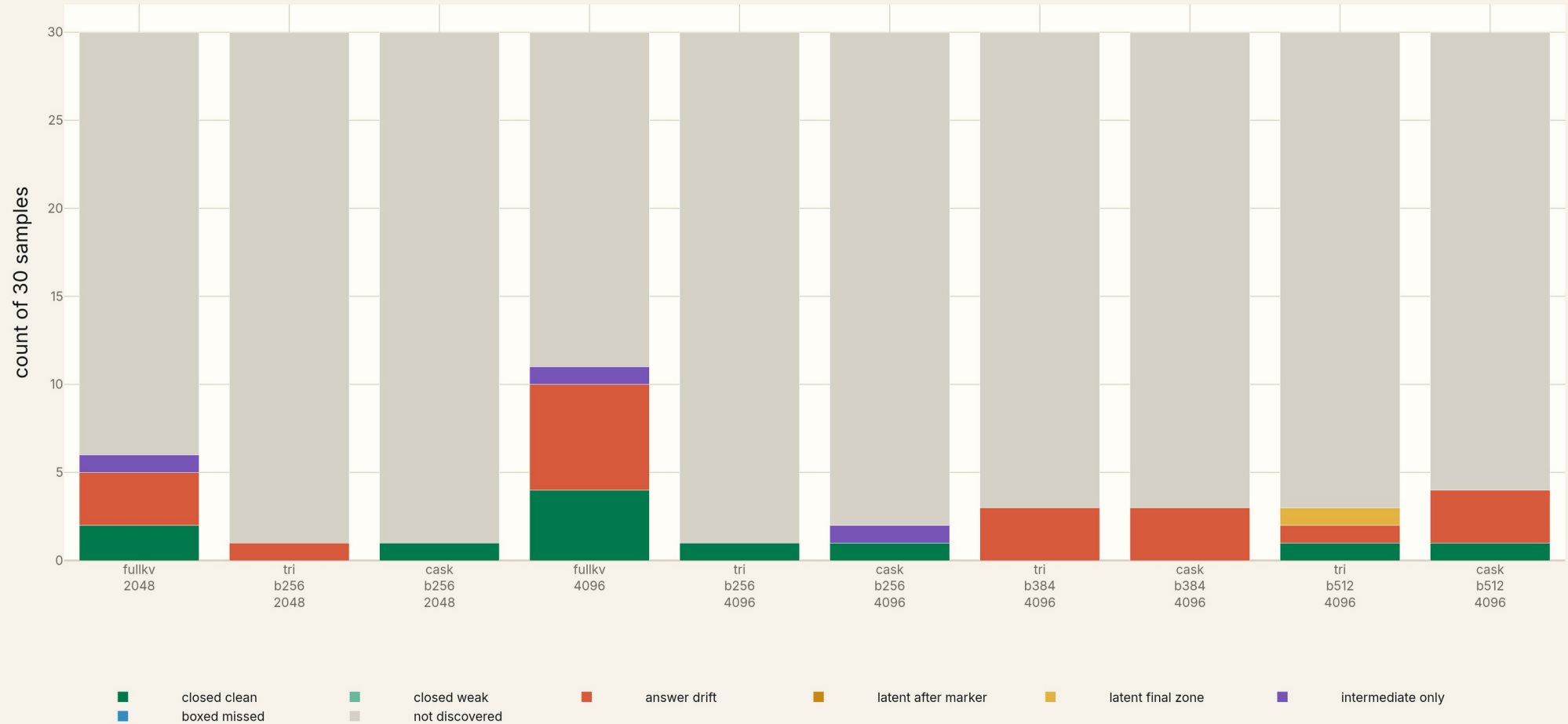
Appendix: Discovery / Retention / Closure counts



Appendix · pathology mix

Raw accuracy collapses these distinct failure modes.

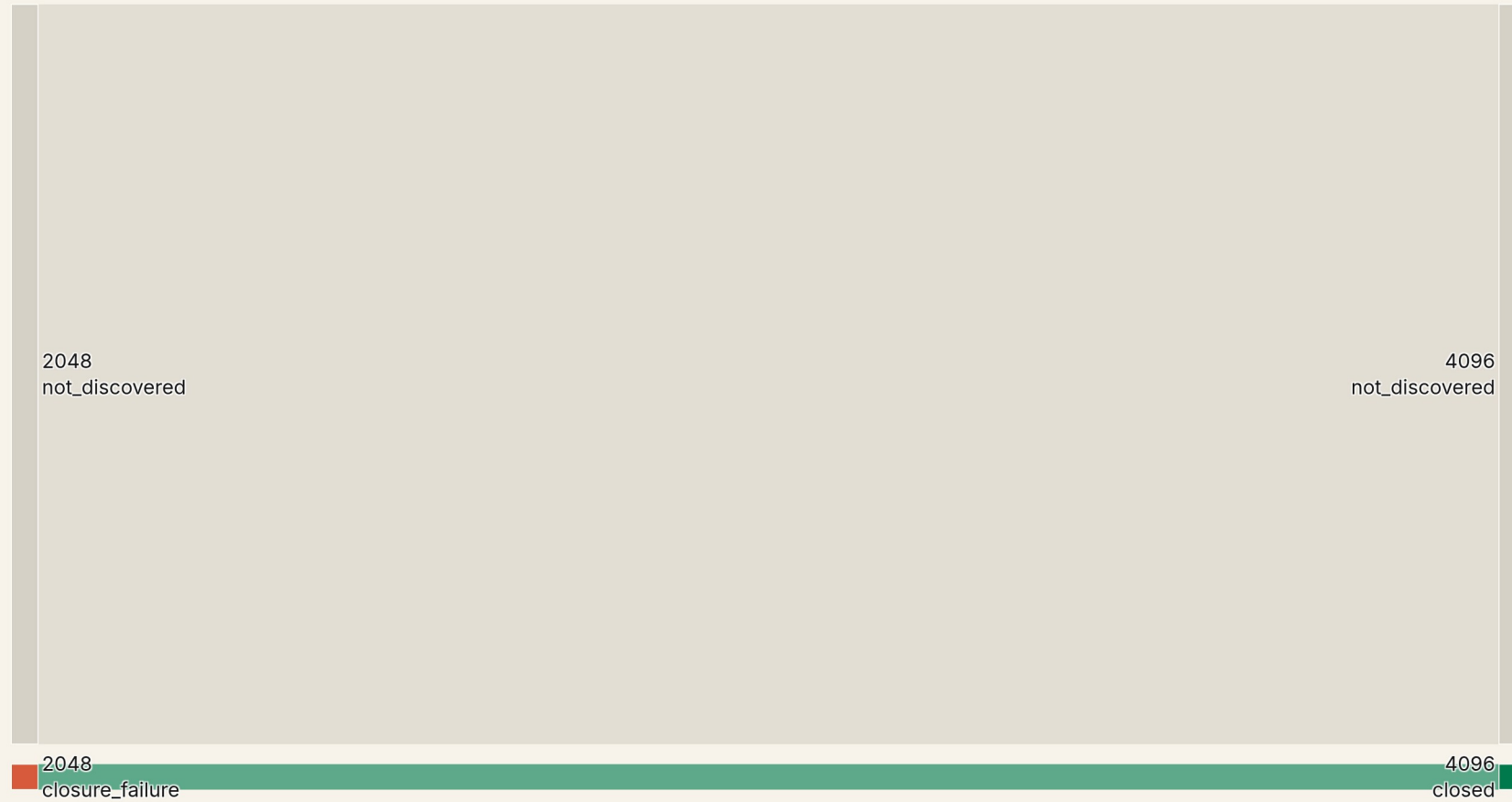
Appendix: pathology mix



Appendix · TriAttention transition

Transition structure for the TriAttention baseline.

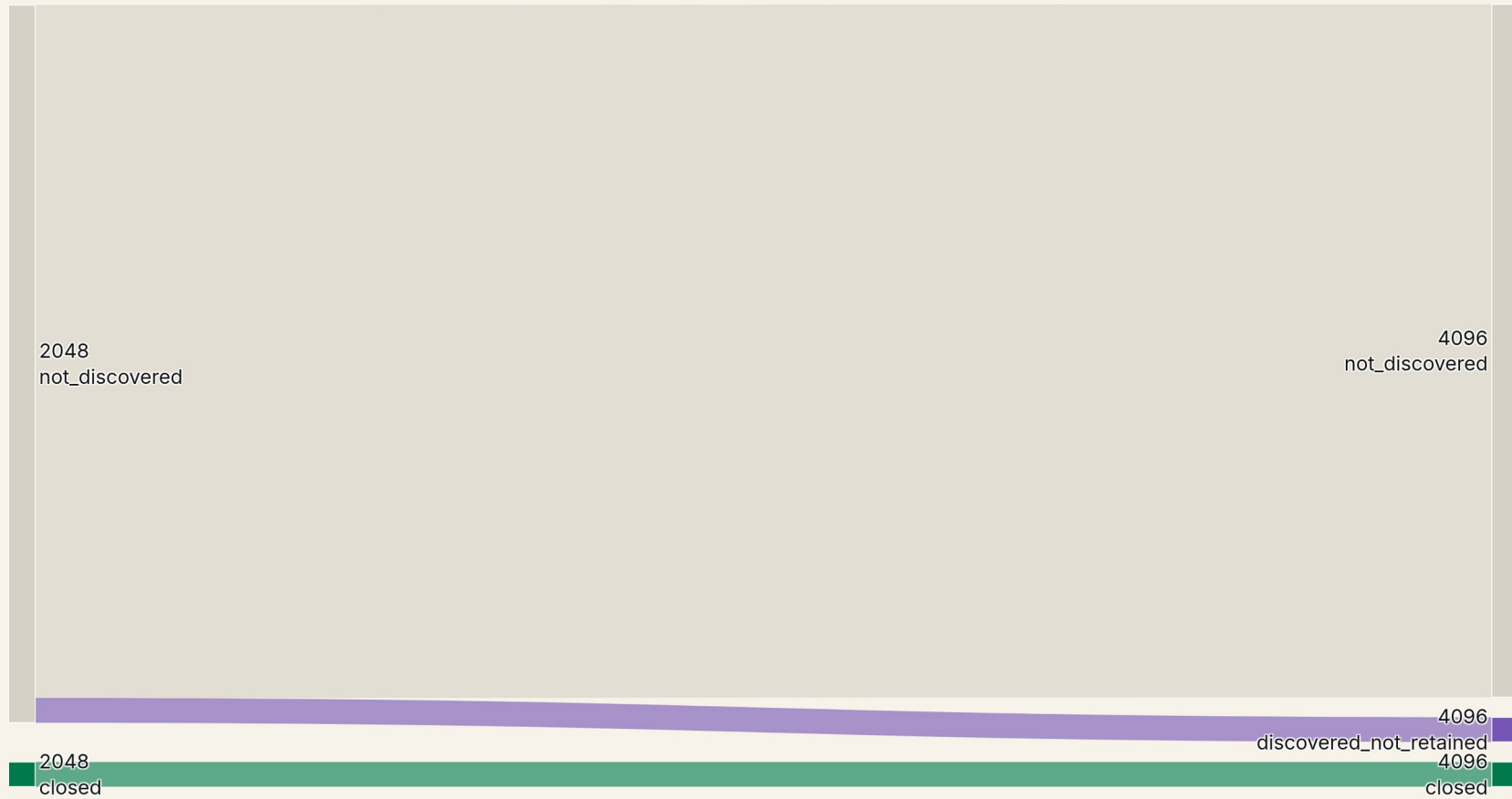
TriAttention b256: 2048 → 4096 transition



Appendix · CASK transition

CASK preserves idx 7, but does not recover FullKV gains here.

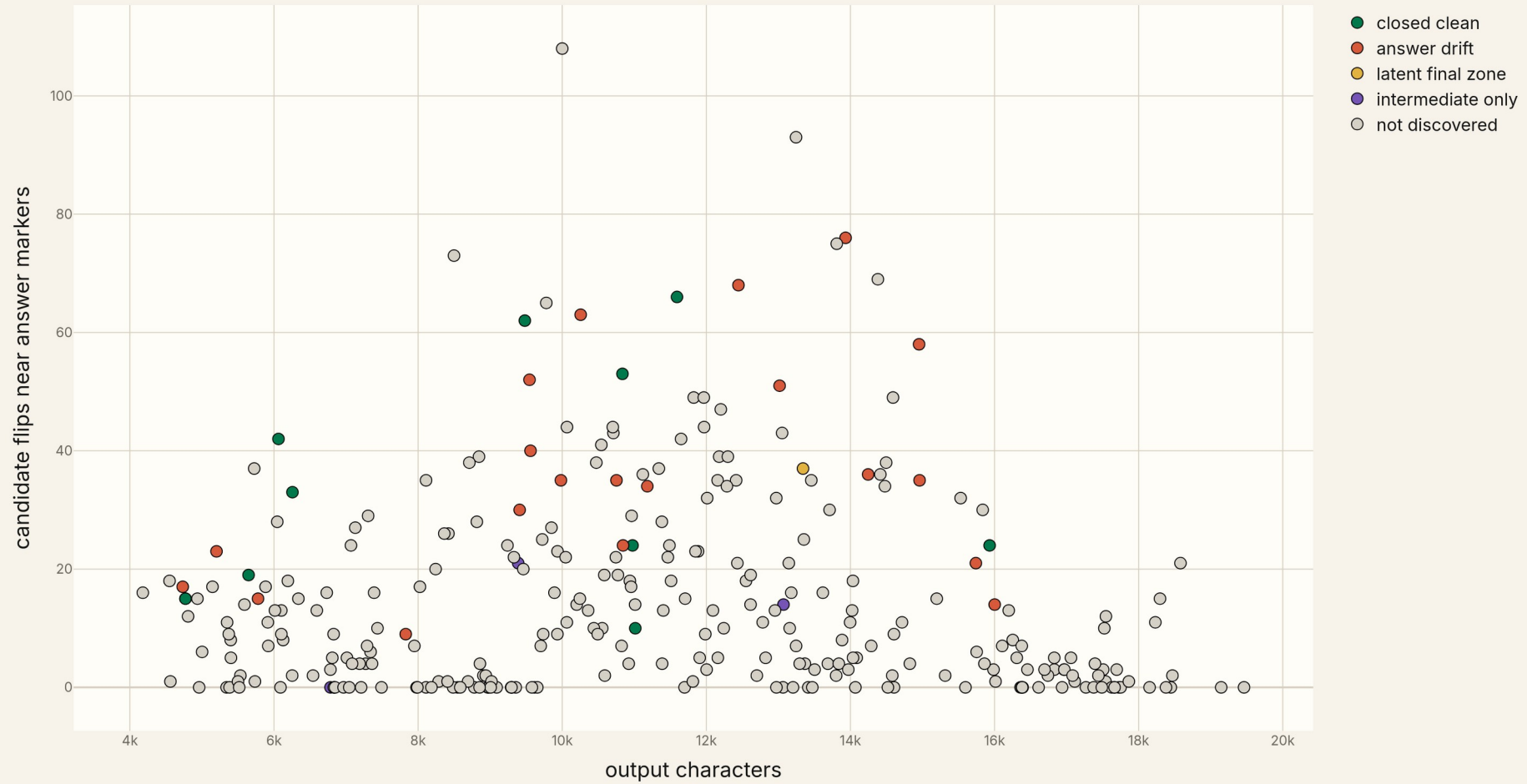
CASK b256: 2048 → 4096 transition



Appendix · candidate churn

Longer generation opens space for candidate drift.

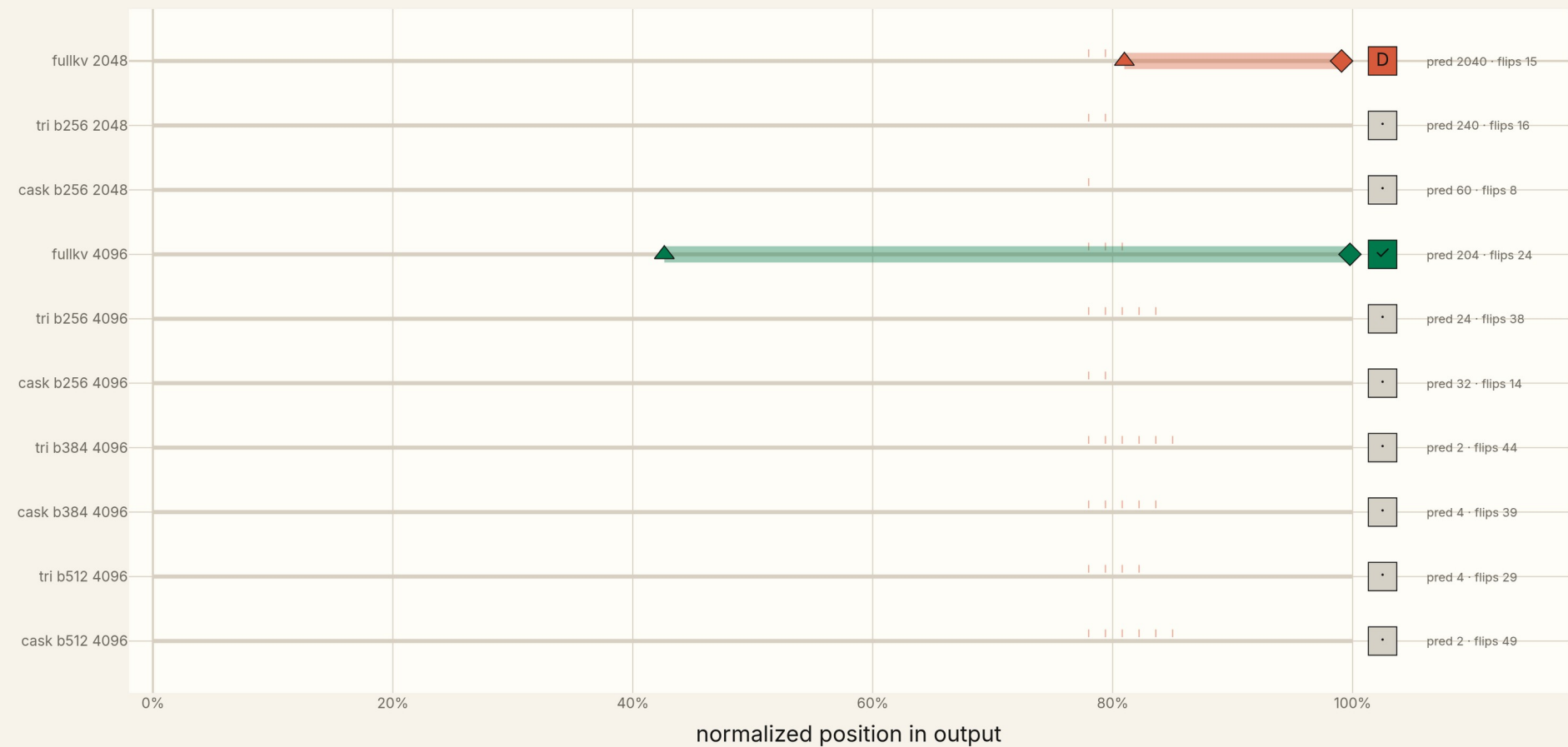
Appendix: candidate churn vs output length



Appendix · idx 0

A FullKV 4096 gain: discovery stabilizes into closure.

Case strip idx 0: GT 204

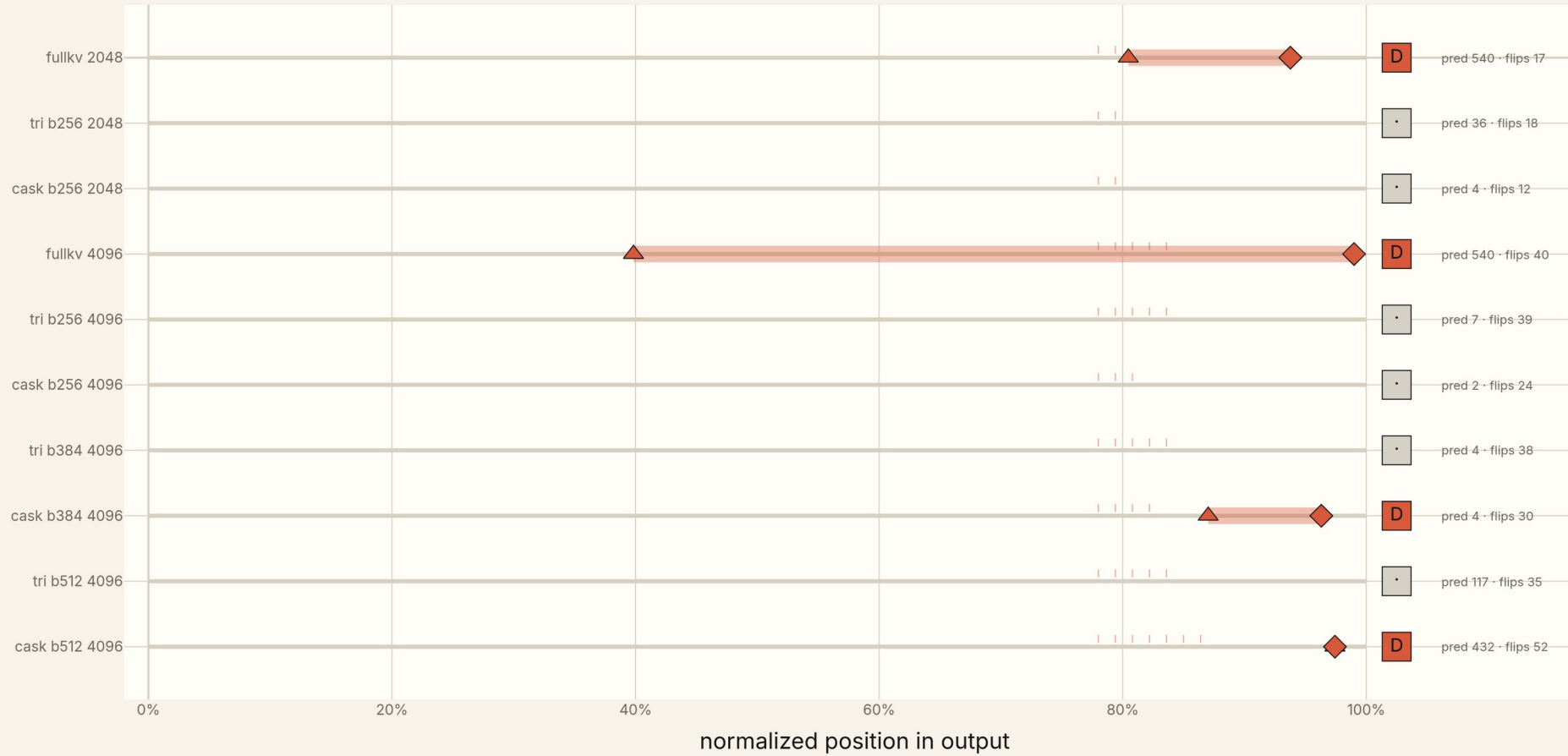


▲ first GT · ◆ last GT · translucent band = GT persistence · orange microticks = candidate churn · square = final pathology

Appendix · idx 12

A drift-heavy case: churn without clean commitment.

Case strip idx 12: GT 540

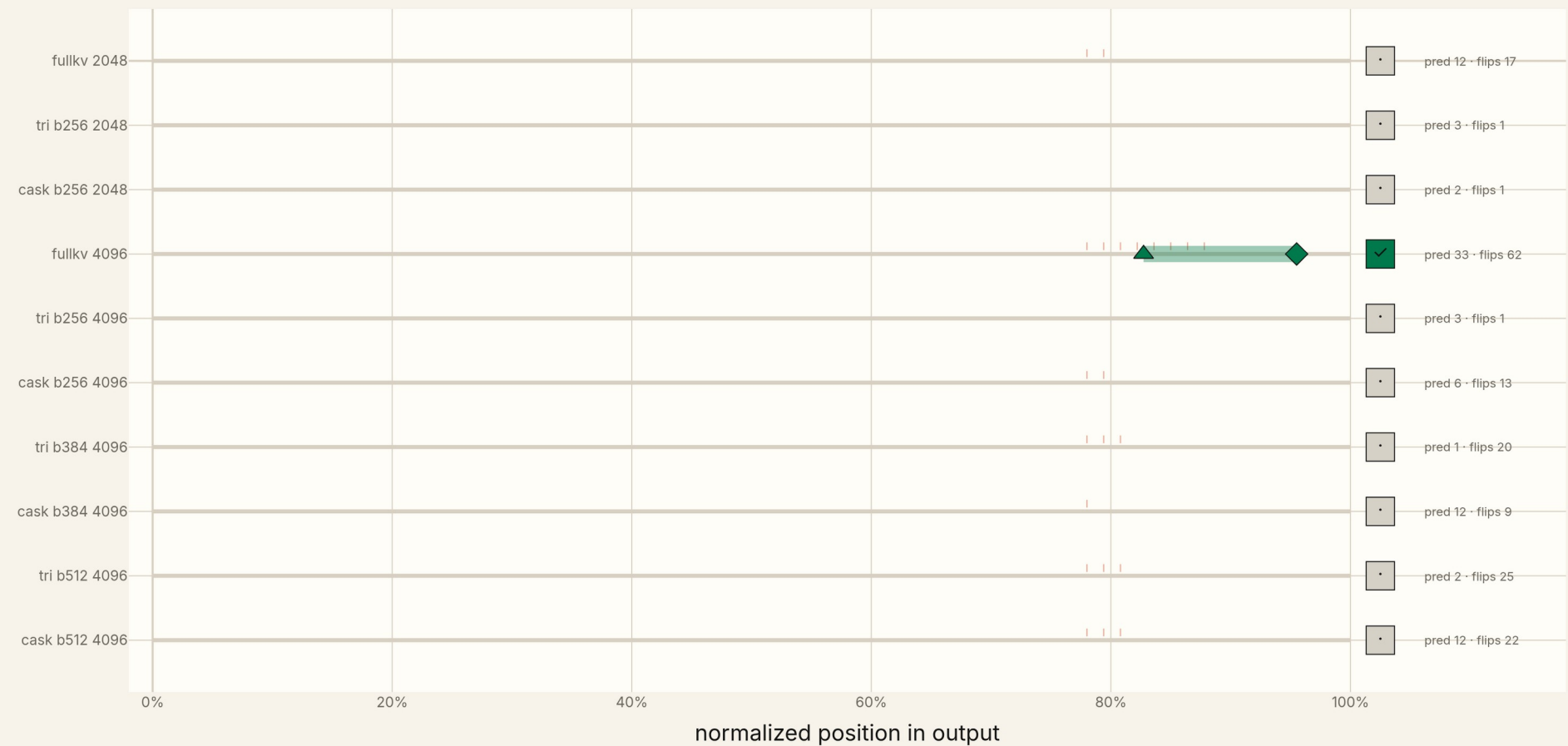


▲ first GT · ◆ last GT · translucent band = GT persistence · orange microticks = candidate churn · square = final pathology

Appendix · idx 24

A FullKV 4096 closure not recovered by compression.

Case strip idx 24: GT 33

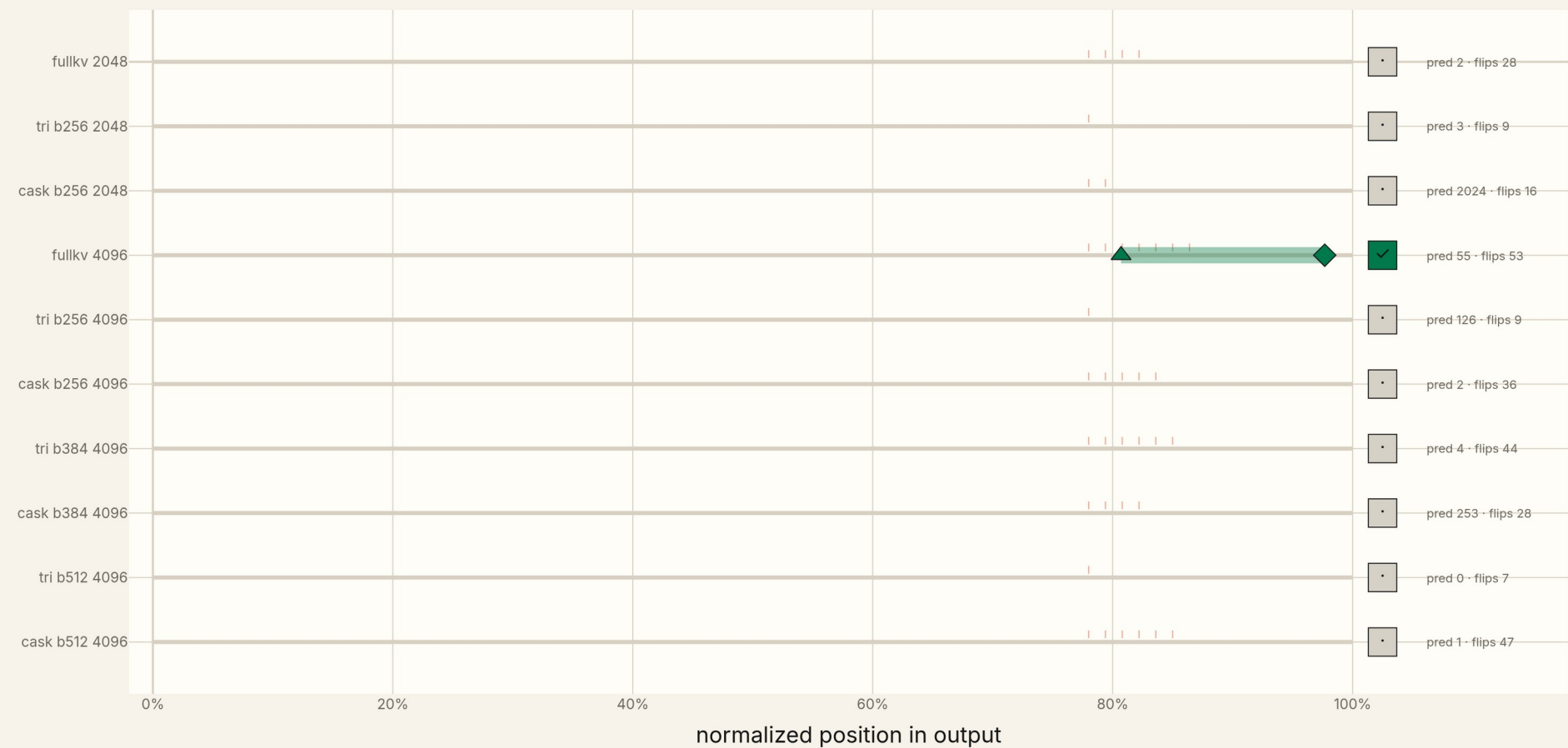


▲ first GT · ◆ last GT · translucent band = GT persistence · orange microticks = candidate churn · square = final pathology

Appendix · idx 26

A FullKV 4096 closure not recovered by compression.

Case strip idx 26: GT 55



▲ first GT · ◆ last GT · translucent band = GT persistence · orange microticks = candidate churn · square = final pathology